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Optimized Shale Gas Fracturing Design Based on Three Sweet Spots Identification: A Case Study from the Southern Sichuan Basin

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Abstract

The marine shale gas reservoirs in the southern Sichuan Basin exhibit strong lateral heterogeneity and complex natural fracture development. Multi-cluster fracturing technology for horizontal wells is an effective method for the efficient development of shale gas. It improves fracturing effectiveness and enhances production potential by adjusting the length of fracturing stages and optimizing fracturing parameters. However, traditional fracturing design methods do not account for differences in reservoir resource abundance or the impact of complex geological structures. This study meticulously divides the marine shale gas reservoir into eight levels based on three factors: reserve enrichment, rock brittleness, and the development degree of natural fractures and bedding. A horizontal well full-section sweet spot identification method, considering the "Geology-Engineering-Fracture Network Construction Potential" (Three Sweet Spots), is proposed. Moreover, targeted reservoir stimulation techniques are assigned to sections with different sweet spot levels. With the goal of maximizing stimulated reservoir volume (SRV), optimal fracturing parameters are calculated for each technique, resulting in a differentiated fracturing design method based on stage and cluster. The "Three Sweet Spots" identification method effectively reveals the distribution characteristics of reservoir-rich areas and regions suitable for fracture network construction along the entire wellbore section. Based on the identification results, stage-cluster parameters are adjusted to optimize the fracturing design. For sweet spot levels 1–4, which have higher resource abundance, the production potential of favorable reservoirs is maximized through strategies such as reducing stage length, increasing Liquid-injected Volume per meter and Sand-injected Volume per meter. For sweet spot levels 5–8, which have lower resource abundance, more economical methods, such as increasing stage length and perforation cluster density, are recommended for the effective development of surrounding reservoirs. The final design well achieved a 7.8% increase in Estimated Ultimate Recovery (EUR) compared to neighboring wells, demonstrating the applicability of the "Three Sweet Spots" identification method in the southern Sichuan region. This paper provides new insights and approaches for optimizing fracturing parameters in horizontal wells by fully considering the heterogeneous distribution

of reservoir resource abundance and rock mechanical properties, offering a new perspective for fracturing design optimization.

Keywords: Shale gas, Sweet spot, Complex geological condition, Fracturing design optimization, Wufeng-Longmaxi shales

Introduction and Background

As conventional oil and gas reserves continue to decline, unconventional resources such as shale gas have entered the stage of commercial development. Horizontal wells combined with large-scale hydraulic fracturing technology have become an effective method for shale gas extraction. In shale reservoirs, the propagation of hydraulic fractures is controlled by various factors, including burial depth (Xie et al., 2021), lithological structure (Sun et al., 2016), in-situ stress distribution (Yong et al., 2022; Zhao et al., 2023), rock mechanical properties (Jansen et al., 2015; Qinghui et al., 1900; Wang et al., 2018), and the development degree of natural fractures and bedding planes (Gale et al., 2014; Gong et al., 2019; Liu et al., 2022b; Yushi et al., 2016; Zhou et al., 2022), all of which ultimately determine the effectiveness of fracturing.

Traditional fracturing parameter optimization methods, based on fracture propagation models such as the finite element method (Hattori et al., 2017), phase field method, displacement discontinuity method (Dehghani Firoozabadi et al., 2022; Xiao and Yue, 2011), and extended finite element method (Chen et al., 2018; Maulianda et al., 2020), aim to maximize the stimulated reservoir volume (SRV) by optimizing construction parameters such as fracture cluster spacing, perforation cluster count, fracturing volume, and fluid injection rate (Yu and Rahman, 2012). However, field case studies have shown that shale reservoirs exhibit heterogeneous characteristics, with uneven distribution of oil and gas resources, differential in-situ stress, and complex natural fracture development (Zhao et al., 2023). Traditional methods do not account for the heterogeneity of shale reservoirs in fracturing design (Jahandideh and Jafarpour, 2016; Li et al., 2012; Lu et al., 2022; Qun et al., 2022).

In the evaluation of shale reservoir heterogeneity, the concept of "sweet spots" has been introduced. Initially, sweet spots referred to high-quality reservoir areas in conventional oil and gas fields. With the advancement of global shale oil and gas development, many scholars have extended this definition to unconventional reservoirs, defining sweet spots as regions with favorable reservoir quality and suitability for fracturing (Bai et al., 2021; Glaser et al., 2013; Wang et al., 2021). Sweet spot identification is a crucial method for assessing shale reservoir quality and guiding fracturing design. Traditional identification methods separately examine geological and engineering sweet spots, primarily considering strength parameters such as shale brittleness and fracture toughness. There is a pressing need for a "sweet spot" identification method that integrates geological and engineering factors (Dongliang, 2020; Liu et al., 2022a; Mezzar et al., 2024; Yang et al., 2025; Zhu et al., 2023; Zou et al., 2019).

The integrated geological-engineering concept, which effectively bridges geological and engineering factors, offers a novel approach to resolving the technical challenges in identifying sweet spots in deep shale gas reservoirs. This study proposes a sweet spot identification method based on the "Geology-Engineering-Fracture Network Construction Potential" model, considering the strong lateral heterogeneity and complex natural fracture development of marine shale gas reservoirs. A fracturing design process suitable for deep shale gas reservoirs is constructed based on this method, which optimizes fracturing parameters by fully considering the heterogeneous distribution of reservoir resource abundance and rock mechanical properties, thus offering new insights and approaches for optimizing horizontal well fracturing design.

Methodology

This study presents a sweet spot identification method that considers the "Geology-Engineering-Fracture Network Construction Potential" (Three Elements) for the full wellbore evaluation of horizontal wells.

Based on this method, a fracturing design process for deep shale gas reservoirs is proposed, as illustrated in Figure 1. The objective of this process is to enhance the Estimated Ultimate Recovery (EUR) of individual wells through differentiated segment management and refined fracturing parameter design. The specific tasks include geological and engineering parameter analysis, Three Elements sweet spot identification, formulation of a differentiated segment strategy based on sweet spot identification, and optimization of fracturing construction parameters based on Stimulated Reservoir Volume (SRV) evaluation. Initially, well log data is used to classify sweet spot levels for the entire horizontal well section using the proposed "Geology-Engineering-Fracture Network Construction Potential" method. Based on the geological and engineering characteristics corresponding to each sweet spot level, appropriate fracturing techniques are proposed and implemented, followed by the differentiation of wellbore sections according to sweet spot distribution. Subsequently, using an SRV calculation model, the multi-fracture propagation patterns under various combinations of horizontal stress differences, discharge, fluid volume, perforation cluster numbers, and cluster spacing are simulated. From these simulations, optimization maps for guiding refined design are developed. Finally, based on the optimization maps, detailed fracturing parameters for each well section subjected to specific fracturing techniques are designed.

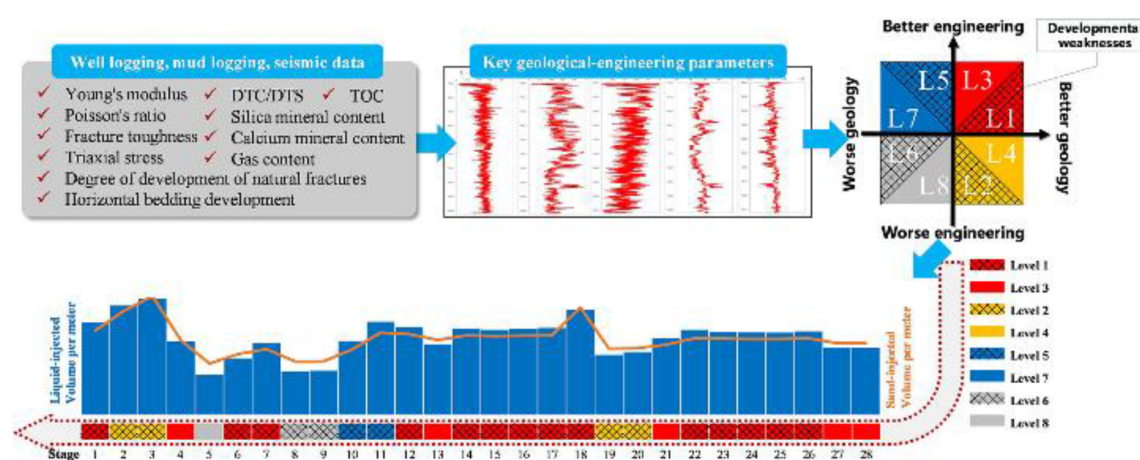


Figure 1—Flow chart of fracturing optimization design based on 8-level sweet spots.

Deep Shale Gas Horizontal Well Full-Section Sweet Spot Identification Method Considering the Three Elements: "Geology-Engineering-Fracture Network Construction Potential"

The southern Sichuan shale gas reservoir is characterized by deep geological burial, complex geological structures, the development of faults and natural fractures, and complex in-situ stress distribution. The ultimate recovery and fracturing effectiveness are influenced by multiple geological and engineering factors. The key factors limiting shale gas production can be categorized into two main aspects: first, the reservoir quality evaluation factors that affect the distribution of resource volume; and second, the rock brittleness evaluation factors that influence the fracturing development effectiveness. Furthermore, the widely developed horizontal bedding planes and microfractures within the shale are conducive to the formation and expansion of the fracture network, thus influencing the ultimate production. Consequently, this study establishes a multi-dimensional sweet spot identification method as follows:

1. **Geological Dimension:** The reservoir quality evaluation model is developed using three parameters: porosity, gas saturation, and Total Organic Carbon (TOC).
2. **Engineering Dimension:** The rock brittleness evaluation model is developed using parameters such as Young's modulus, Poisson's ratio, triaxial in-situ stress, and horizontal stress difference.
3. **Fracture Network Construction Potential Dimension:** The mineral content method is used to calculate the strength of horizontal bedding and natural fracture development, quantitatively

characterizing the degree of shale bedding and microfracture development, thereby establishing the fracture network construction potential evaluation model.

Ultimately, a full-section sweet spot identification method for deep shale gas horizontal wells is proposed, considering the three elements: "Geology-Engineering-Fracture Network Construction Potential."

Reservoir Quality Evaluation Model. Reservoir quality directly influences the recoverable reserves of a reservoir. Conventional gas reservoirs are typically evaluated based on physical properties such as porosity, permeability, and gas saturation. However, shale gas reservoirs, due to their low porosity and low permeability characteristics, derive their recoverable reserves mainly from free gas. Among these, porosity and gas saturation serve as indicators of free gas content. The Total Organic Carbon (TOC) content reflects the abundance of organic matter in shale, which, as a source rock, will convert into both free gas and adsorbed gas. Adsorbed gas, as an important supplement to free gas, can desorb and be produced as free gas during the extraction process as reservoir pressure decreases (Li et al., 2025). Based on these mechanisms, this study selects porosity, gas saturation, and TOC to establish a reservoir quality evaluation index. The entropy weight method is employed to determine the parameter weights, enabling a quantitative evaluation of the shale reservoir's physical properties. The index expression is as follows:

$$\begin{cases} B_G = \alpha TOC_N + \beta V_{fN} \\ V_{fN} = S_{gN} \times \phi_N \end{cases}$$

Where: ϕ_N , S_{gN} , TOC_N represent the normalized well log data for porosity, gas saturation, and TOC, respectively. To establish a quantitative relationship between the three parameters and the reservoir quality evaluation index, the entropy weight method is applied to determine the weight coefficients α and β .

Rock Brittleness Evaluation.

Rock Brittleness Characterization Coefficient

Rock matrix brittleness represents its overall mechanical behavior, expressed as the ability of the rock to form a network of fractures under external loading. High-brittleness rocks are characterized by internal shear deformation occurring at relatively low loads, generating numerous fracture surfaces. From a mechanical perspective, Poisson's ratio (ν) reflects the rock's resistance to compressive failure, while Young's modulus (E) characterizes the rock's ability to maintain fractures after rupture. Brittleness is positively correlated with Young's modulus and negatively correlated with Poisson's ratio (Zhang et al., 2016). From a mineralogical perspective, rocks with a higher content of brittle minerals (such as calcite and quartz) exhibit greater brittleness. Calcareous/siliciclastic minerals show stronger resistance to deformation in sedimentary strata (Zhang et al., 2017). Therefore, a rock brittleness characterization coefficient is established using a combination of mechanical parameters and mineral composition. The mineral content supplements the mechanical parameters by reflecting the rock's failure process at low strain, which cannot be captured by mechanical parameters alone. In turn, the mechanical parameters correct for the changes in the mechanical properties of minerals during the sedimentary diagenesis process. The rock brittleness characterization coefficient is calculated as follows:

$$F_m = \frac{B_e + B_m}{2} \quad (1)$$

In the equation:

$$\begin{cases} B_e = E_N / \nu_N \\ B_m = \frac{W_{\text{quartz}} + W_{\text{cal}}}{W_{\text{quartz}} + W_{\text{cal}} + W_{\text{mud}}} \times 100\% \end{cases} \quad (2)$$

Where: B_e , B_m represent the rock mechanical brittleness and rock mineral brittleness, respectively; E_N , ν_N represent the normalized values of Young's modulus and Poisson's ratio, respectively; W_{quartz} , W_{cal} , W_{mud} represent the contents of siliceous minerals, carbonate minerals, and clay minerals, respectively.

In-Situ Stress Difference Coefficient

In-situ stress is a key controlling factor in shale reservoir fracturing, governing the fracturing effectiveness through a dual mechanism. On one hand, the horizontal stress difference directly regulates the complexity of the fracture network: under high stress difference conditions, a dominant main fracture is more likely to form, while a low stress difference environment promotes the development of a more complex fracture network. On the other hand, the minimum horizontal principal stress shows a significant positive correlation with the rock fracture pressure; in low-stress reservoirs, fractures are more likely to initiate (Zhou et al., 2023). Based on the aforementioned in-situ stress control mechanisms, the in-situ stress difference coefficient is established, and its calculation formula is as follows:

$$S_d = \frac{|S_{h\min} - S_v|}{S_{H\max} - S_{h\min}} \quad (3)$$

Where: S_d represents the in-situ stress difference coefficient; $S_{H\max}$, $S_{h\min}$, S_v represent the three principal stresses, respectively.

The shale matrix fracture network construction capacity is controlled by mineral composition, elastic mechanical parameters, and in-situ stress distribution characteristics. The rock brittleness characterization coefficient and in-situ stress difference coefficient are generally positively correlated with the degree of fracture network development. Based on this, the rock stress brittleness index is constructed, and its expression is as follows:

$$B_E = aF_M + bS_D \quad (4)$$

Where: B_E represents the rock stress brittleness index; F_M , S_D represent the normalized values of the rock brittleness characterization coefficient and in-situ stress difference coefficient, respectively; a and b are regional adaptability weighting coefficients, typically within the range: $a \in [0.23, 0.34]$, $b \in [0.66, 0.77]$; for this block, the values are: $a = 0.34$, $b = 0.66$.

Fracture Network Construction Potential Evaluation. The core difference between shale and conventional reservoirs lies in the development of numerous microscopic natural fractures (structural fractures/bedding fractures/slip fractures), which exhibit a dual function. On one hand, these fractures enhance hydrocarbon recoverability by providing pathways for the desorption of adsorbed gas and storage space for free gas. On the other hand, the development of fractures and the weak cementation characteristics of bedding significantly reduce the shale's fracture toughness, facilitating fluid infiltration and expanding the coverage area of hydraulic fractures. Research on the primary mechanisms controlling fracture development has shown that quartz content is a key factor influencing fracture formation. Black shale, which is rich in quartz, is more brittle and has a higher density of natural fractures compared to gray shale rich in calcite. Additionally, the natural weak planes are controlled by the joint effects of biogenic silica enrichment, diagenetic shrinkage, and hydrocarbon overpressure, which are significantly positively correlated with organic matter and siliceous components. Based on Shen Cheng's mineral regression model (Shen Cheng et al., 2019), a regression equation with siliceous mineral content and carbonate mineral content as independent variables is established, and the fracture network construction potential index is derived as follows:

$$B_{NF} = \frac{2(S_N \times I_N)}{S_N + I_N} \quad (5)$$

$$\text{Natural Fracture Development Index: } I_n = 0.5 \times W_{quartz} + 2 \times W_{cal} + 55.0 \quad (6)$$

$$\text{Horizontal Bedding Development Index: } S_n = \begin{cases} 5.5 \times W_{\text{quartz}} - 130.0, & W_{\text{quartz}} < 50 \\ -W_{\text{quartz}} + 1.5 \times W_{\text{cal}} + 150, & W_{\text{quartz}} > 50 \end{cases} \quad (7)$$

Where: B_{NF} represents the fracture network construction potential index; I_{N} and S_{N} represent the normalized values of the natural fracture development index and horizontal bedding development index, respectively.

Differentiated Fracturing Design Method Based on the "Geology-Engineering-Fracture Network Construction Potential" Three Sweet Spot Identification Method

The core of economically efficient fracturing for shale reservoirs lies in the full stimulation of high-quality geological sweet spots. Based on this, a graded stimulation strategy is proposed in this study. First, the sweet spot level for the target horizontal well section is classified based on the comparison of sweet spot index results along the wellbore with the geological, engineering, and fracture network potential thresholds. Next, fracturing construction techniques tailored to the block are recommended, considering the resource characteristics, fracture network construction difficulty, and regional operational experience corresponding to each sweet spot level. This strategy aims to achieve the synergistic goal of optimizing technical conditions and maximizing economic benefits, ultimately enhancing the Estimated Ultimate Recovery (EUR) of the well. The criteria for sweet spot classification and the corresponding fracturing process recommendations are detailed in Table 1.

Table 1—Shale Sweet Spot Index Classification and Fracturing Technique Recommendations.

Sweet Spot Level	Reservoir Quality Evaluation Index (BG)	Rock Brittleness Index (BE)	Fracture Network Construction Potential Index (BNF)	Fracturing Technique Recommendations
1	$B_G > B_G^{\text{crit}}$	$B_E > B_E^{\text{crit}}$	$B_{\text{NF}} > B_{\text{NF}}^{\text{crit}}$	Medium-stage, moderate-cluster, moderate-plugging
2	$B_G > B_G^{\text{crit}}$	$B_E < B_E^{\text{crit}}$	$B_{\text{NF}} > B_{\text{NF}}^{\text{crit}}$	Short-stage, low-cluster, high-plugging
3	$B_G > B_G^{\text{crit}}$	$B_E > B_E^{\text{crit}}$	$B_{\text{NF}} < B_{\text{NF}}^{\text{crit}}$	Medium-stage, high-cluster, low-plugging – tight cut
4	$B_G > B_G^{\text{crit}}$	$B_E < B_E^{\text{crit}}$	$B_{\text{NF}} < B_{\text{NF}}^{\text{crit}}$	Medium-stage, high-cluster, no plugging – tight cut
5	$B_G < B_G^{\text{crit}}$	$B_E > B_E^{\text{crit}}$	$B_{\text{NF}} > B_{\text{NF}}^{\text{crit}}$	Long-stage, moderate-cluster, moderate-plugging
6	$B_G < B_G^{\text{crit}}$	$B_E < B_E^{\text{crit}}$	$B_{\text{NF}} > B_{\text{NF}}^{\text{crit}}$	Long-stage, moderate-cluster, high-plugging
7	$B_G < B_G^{\text{crit}}$	$B_E > B_E^{\text{crit}}$	$B_{\text{NF}} < B_{\text{NF}}^{\text{crit}}$	Extra-long-stage, moderate-cluster, low-plugging – tight cut
8	$B_G < B_G^{\text{crit}}$	$B_E < B_E^{\text{crit}}$	$B_{\text{NF}} < B_{\text{NF}}^{\text{crit}}$	Extra-long-stage, moderate-cluster, no plugging – tight cut

Differentiated Stimulation Strategy and Targeted Fracturing Parameter Optimization Based on Sweet Spot Classification

The reservoir quality evaluation index characterizes the resource abundance of the reservoir and is a key factor in determining the Estimated Ultimate Recovery (EUR) of a well. For different resource abundances, the following optimization strategies are adopted:

For High Resource Abundance Reservoirs (Levels 1–4): Reduce Stage Length: Shorten stage lengths to improve the precision of stimulation in individual stages, ensuring fracturing energy is concentrated in the most favorable reservoir zones. Moderate Perforation Cluster Count: Optimize the number of perforation

clusters to enhance uniformity and fracture network complexity. Increase Fluid and Sand Injection Strength: Enhance fracture extension and proppant support capacity. Use Temporary Plugs Appropriately: Prevent fluid migration in high-quality areas and increase the Stimulated Reservoir Volume (SRV) by fully utilizing the high-quality reservoir. This strategy maximizes the productivity potential of high-quality reservoirs, ensuring sustained production growth.

For Low Resource Abundance Reservoirs (Levels 5–8): Increase Stage Length: Extend stage lengths to broaden the range of stimulation in each stage and reduce the number of stages to decrease construction costs. Increase Perforation Cluster Density: Increase cluster density to improve the probability of fracture initiation, compensating for the reservoir's inherent limitations. Reduce Fluid and Sand Injection Strength: Control material consumption to enhance the economic feasibility of stimulation. Minimize or Avoid Temporary Plugs: Simplify operations and optimize the fracturing process. This strategy ensures effective utilization of surrounding resources in low-resource zones while optimizing the overall economic benefits of development.

Rock Brittleness Index primarily reflects the rock's mechanical properties, brittleness, and stress state, which are direct factors in the ease of fracture initiation and propagation during fracturing. For different rock brittleness indices, the following optimization strategies are proposed:

For High Rock Brittleness Index Reservoirs (Levels 1, 3, 5, 7): Easier Fracture Initiation: Use medium to high-strength fracturing fluid systems to ensure efficient fracture propagation. Control Injection Rate: Prevent excessive extension of fractures that could cause fluid migration. These measures fully release the natural stimulation potential of high-brittleness areas, ensuring efficient use of fracturing energy.

For Low Rock Brittleness Index Reservoirs (Levels 2, 4, 6, 8): Difficult Fracture Development: Use high-flow-rate fracturing techniques to increase pressure gradients and facilitate fracture initiation and propagation. Adjust Fluid System Viscosity and Sand Strength: Tailor the fracturing fluid system to ensure sufficient fracture support and formation of a fracture network. This strategy enhances the fracturing efficiency in low-brittleness areas, increasing the micro-stimulation volume.

Fracture Network Construction Potential Index reflects the degree of development of natural fractures and bedding structures in the reservoir, which is critical for the formation of a complex fracture network post-fracturing. For different fracture network construction potential indices, the following optimization strategies are adopted:

For High Fracture Network Construction Potential Index (Levels 1, 2, 5, 6): Natural Fracture Activation: Use appropriate temporary plugging agents to avoid excessive fracture propagation in dominant fractures. Increase Sand Injection Strength: Enhance fracture support and maintain the continuity of fracture conduits. Increase Injection Rate: Accelerate pressure transmission to facilitate fracture connectivity. These measures effectively activate existing micro-fractures in the reservoir, creating a high-complexity SRV and boosting production capacity.

For Low Fracture Network Construction Potential Index (Levels 3, 4, 7, 8): Limited Natural Fractures: Rely on artificial fracturing to construct the fracture network. Dense Perforation Cluster Design: Increase the perforation cluster density to create more fractures. High-Strength Fluid Systems and High-Flow Techniques: Enhance the number of primary and branch fractures. Extend Stage Lengths: Concentrate energy and increase the artificial stimulation range. Optimize Temporary Plugging Agents: Ensure frequent branching of primary fractures to improve the artificial fracture network complexity. Adjust Sanding Program: Ensure sufficient support strength for fractures after formation to prevent fracture closure. This strategy compensates for the lack of natural fracture networks by using artificial techniques to construct complex fracture networks.

Case Study

A case study is presented using the LU-X-1H well to demonstrate the application of the Three Sweet Spot Prediction and Differentiated Fracturing Design.

The LU-X-1H well is located in the X wellblock of the Luzhou region in the Sichuan Basin. The target reservoir is a black shale rich in organic matter, formed in a deep-water shelf depositional environment, with the main production intervals being the Wufeng Formation and Longmaxi Formation. The area is structurally complex, with a dense interwoven system of low-angle bedding fractures and high-angle structural fractures, forming a network of microfractures. The wellbore crosses multiple faults and major fracture zones, resulting in significant reservoir heterogeneity.

The horizontal section of the well is located between 3814.00 m and 5479.00 m, with a total length of 1665 m. The reservoir petrophysical parameters indicate an average total porosity of 4.65%, an average total organic carbon content (TOC) of 3.88%, and an average total gas content of 7.19 cm³/g. As shown in [Figures 2\(a\)–\(d\)](#), the distribution of petrophysical parameters (porosity, TOC, gas content) along the horizontal section (from the heel to the toe) is uneven, with high values concentrated in four sections: 3814–4100 m, 4167–4226 m, 4500–4620 m, and 4790–5266 m. Regarding rock mechanical properties, the average Young's modulus along the horizontal section is 40.57 MPa, and the average Poisson's ratio is 0.23. [Figures 2\(e\)–\(g\)](#) reveal that the intervals from 3814–4200 m, 4529–4723 m, and 4821–5326 m exhibit relatively high Young's modulus and low Poisson's ratio, indicating that these sections are more brittle and conducive to fracturing. Additionally, the minimum principal stress values in these sections are lower, making them more prone to fracture initiation. Furthermore, [Figure 2\(h\)](#) shows that the 3850–4230 m and 4820–5256 m intervals have a higher degree of natural fracture development, and the bedding is relatively uniform across the entire wellbore section, which plays a crucial role in the formation of a complex fracture network during the fracturing process.

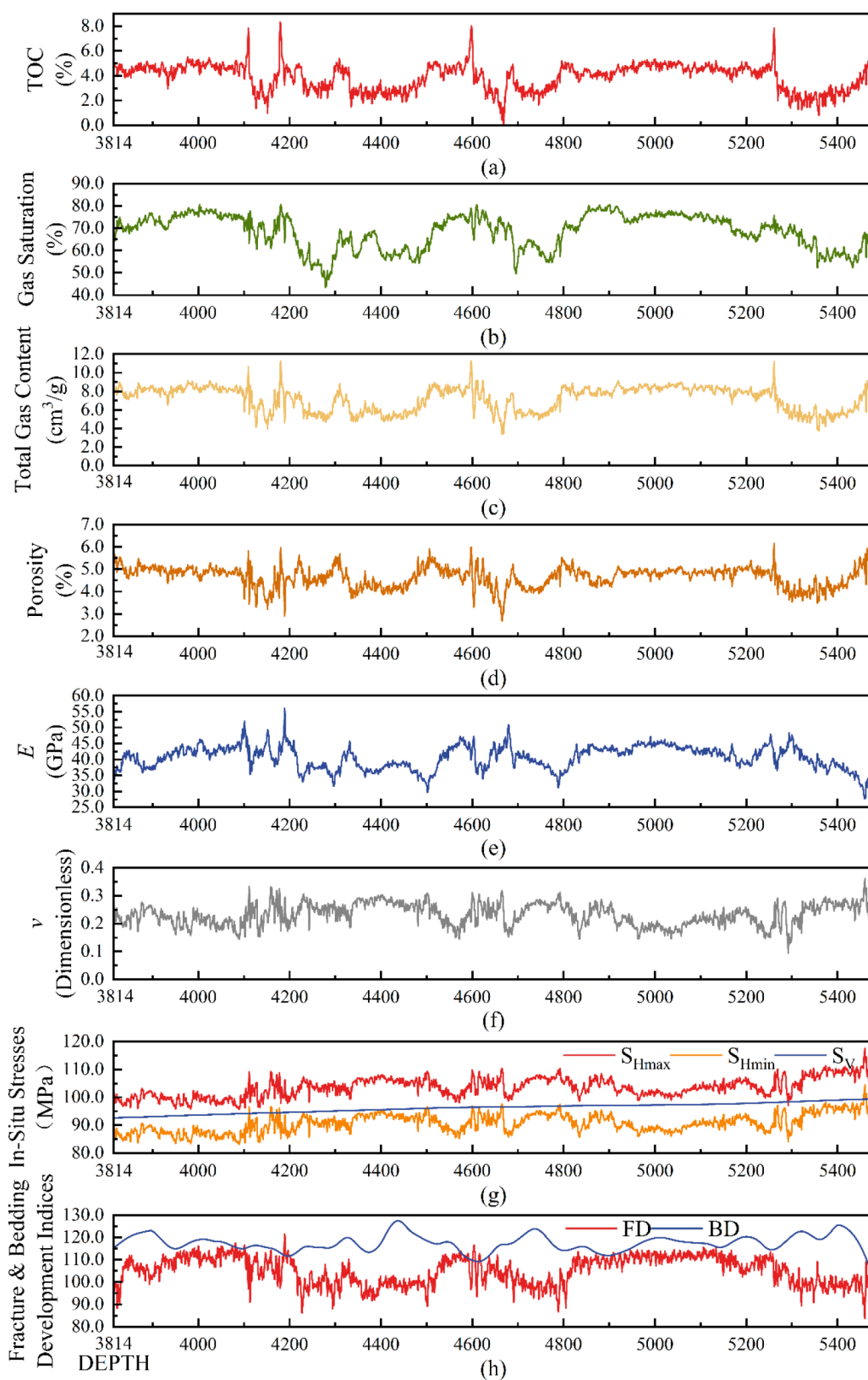


Figure 2—Distribution along the LU-X-1H Horizontal Section: (a) TOC; (b) Gas Saturation; (c) Total Gas Content; (d) Porosity; (e) Young's Modulus; (f) Poisson's Ratio; (g) In-Situ Stresses; (h) Fracture & Bedding Development Indexes.

Based on the key indicators from the geological, engineering, and fracture network development potential aspects, a comprehensive evaluation and classification of the entire wellbore section of the LU-X-1H well

were conducted. The final distribution of the "Geology-Engineering-Fracture Network Potential" Three Sweet Spot Levels is shown in Figure 3.

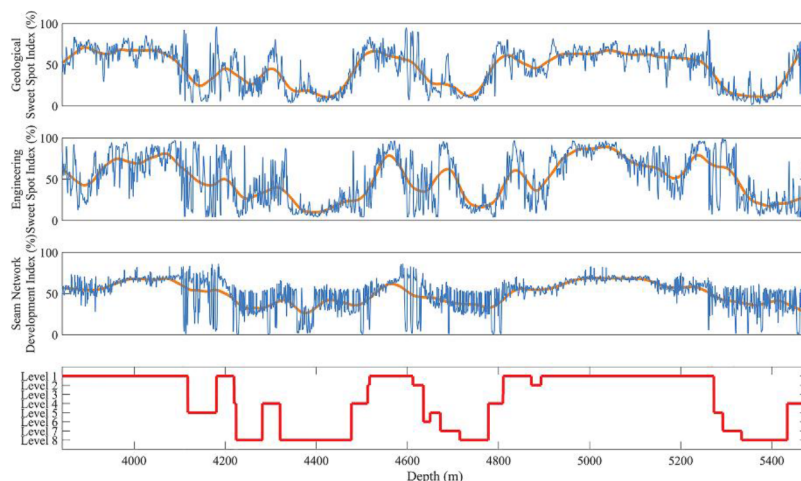


Figure 3—Calculation of sweet spot classification in horizontal well sections.

Based on the comprehensive evaluation results of the Three Sweet Spots, a 19-stage differentiated fracturing design was implemented for the LU-X-1H well. The evaluation results, as shown in Figure 4, indicate significant variations in the reservoir quality index, rock brittleness index, and fracture network construction potential index along the wellbore. Furthermore, within each stage, there are varying degrees of deviation in the individual indices. Based on the integrated evaluation of each stage, targeted fracturing techniques were developed for each sweet spot level, aiming to match the heterogeneity of the reservoir and fully release the reservoir's potential, thereby maximizing economic benefits. The relationship between the single-stage sweet spot level, fluid injection strength, and production capacity for the LU-X-1H well is shown in Figure 5.

The 3rd–9th, 12th, and 17th–19th stages are classified as Sweet Spot Level 1 (Main Production Stages), with substantial recoverable resource potential and good fracturability. For these stages, a medium-stage length, moderate perforation cluster count, and moderate temporary plugging process were used to effectively form a fracture network with strong connectivity, conducive to hydrocarbon migration. These stages exhibit relatively easy fracture network construction, with overall production significantly higher than other stages. The production from a single stage is mainly concentrated between 40,000 to 98,000 m³/day, with Stage 8 producing 83,000 m³/day and Stage 19 reaching 98,000 m³/day, the highest across the entire well.

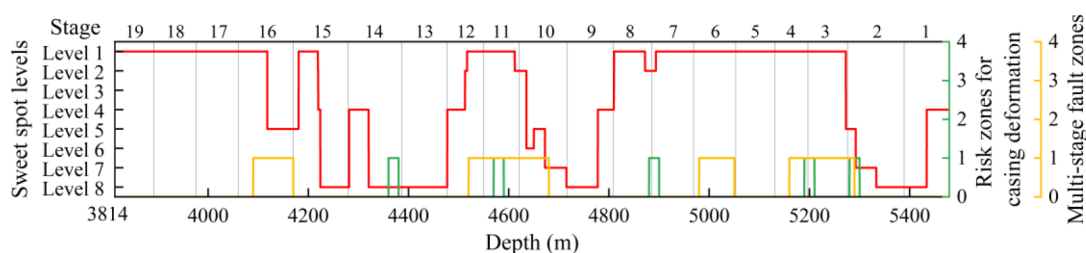


Figure 4—Staging design of LU-X-1H horizontal well based on Class 8 sweet spot classification.

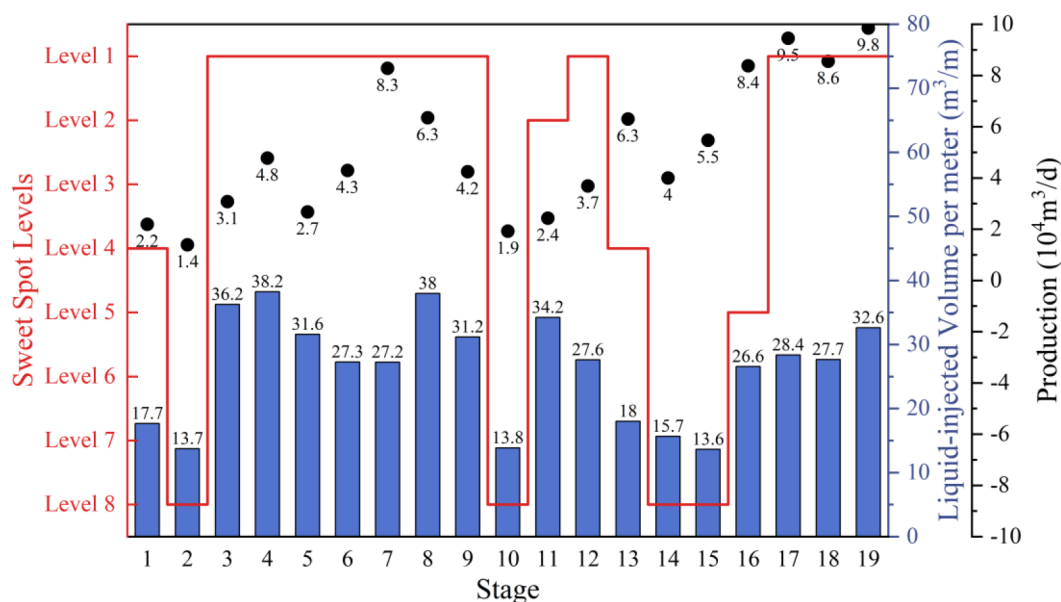


Figure 5—Comparison of Sweet Spot Classification, Liquid-injected Volume per meter, and Production across Stages in LU-X-1H Horizontal Well.

The 11th stage is classified as Sweet Spot Level 2, with abundant resources and high rock brittleness, but weak microfracture and bedding development, limiting fracture network construction potential. Its production is 37,000 m³/day, placing it at a moderate level. For this stage, a short stage, low perforation cluster count, and high temporary plugging process were used, with combined control of fracture mouth and fracture internal plugging. This approach ensures uniform fracture propagation, facilitates fracture reorientation, and enhances fracture complexity and stimulation coverage.

The 1st and 13th stages are classified as Sweet Spot Level 4, with certain recoverable resources but high rock plasticity and weak natural fractures and bedding development, which limits fracture conductivity. Their production is 22,000 and 19,000 m³/day, respectively, approaching the lowest values in the well. To improve stimulation effectiveness, a short stage, high perforation cluster count, and no temporary plugging – tight cut process were used. By increasing cluster density and shortening stage length, multiple fractures were induced, achieving effective communication of shale oil and gas.

The 16th stage is classified as Sweet Spot Level 5, with relatively small resource volumes but good rock brittleness and well-developed microfractures and bedding. The production from this stage is 55,000 m³/day, which is at a mid-low level. This stage adopted a long stage, high perforation cluster count, and moderate temporary plugging process, controlling fracture propagation to balance fracture network construction and construction economy.

The 2nd, 10th, and 14th–15th stages are classified as Sweet Spot Level 8, with limited recoverable resources and significant challenges in fracture network construction. Production is generally low, such as 14,000 m³/day for Stage 2, 24,000 m³/day for Stage 14, and 40,000 m³/day for Stage 15. A super-long stage, ultra-high perforation cluster count, and no temporary plugging process were adopted, primarily to relieve blockage in the near-wellbore zone, reduce construction costs, and optimize the stimulation effectiveness.

Overall, there is a clear positive correlation between sweet spot level and production: the higher the sweet spot level, the stronger the fracturability, the easier the fracture construction, and the higher the production; conversely, the lower the sweet spot level, the poorer the fracturability, and the more significantly reduced the production.

For the other three wells on the LU-X platform, conventional fracturing designs were used. After two years of production, the comparison of capacity is shown in Figure 6: the EUR per kilometer of the LU-

X-1H well reached $0.69 \times 10^8 \text{ m}^3$, an improvement of 7.8% compared to the average value of the neighboring wells on the same platform.

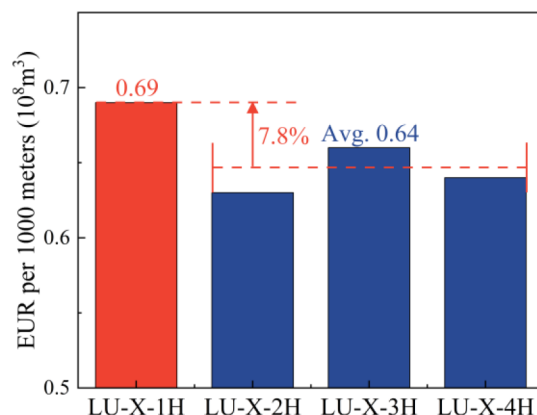


Figure 6—Comparison of EUR per Kilometer in LU-X-1H vs. Offset Wells.

Conclusions

This paper proposes a comprehensive sweet spot identification method for the entire wellbore section, integrating the three elements of "Geology-Engineering-Fracture Network Construction Potential," specifically designed to address the heterogeneity characteristics of marine shale gas reservoirs in the southern Sichuan region. Based on this method, a differentiated fracturing design process has been established. The following insights were obtained:

1. By classifying the reservoir into eight sweet spot levels, this method fully considers the combined impact of reservoir quality, rock brittleness, and fracture network construction potential, and develops targeted strategies for segmenting, clustering, and optimizing fracturing parameters.
2. The method was applied to the LU-X-1H well in a field trial, where a 19-stage differentiated fracturing design was implemented. The results show that the Three Sweet Spot identification and differentiated fracturing design effectively match the heterogeneity of the reservoir, ensuring full stimulation of the advantageous zones and economical utilization of the less favorable zones.
3. Field data show a significant positive correlation between sweet spot levels and single-stage production. The EUR of the LU-X-1H well increased by 7.8% compared to neighboring wells, validating the effectiveness and applicability of the Three Sweet Spot identification and differentiated fracturing design method.

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